EXPERIMENT-24

24. RSA System – C Program to Find the Private Key

AIM

To write a C program to determine the private key d of an RSA system when the public key is $e=31$ and $n=3599$, using trial-and-error factorization and the Extended Euclidean Algorithm.

PROCEDURE

1. Read the public key values $e=31$ and $n=3599$.
2. Find the prime factors p and q of n using trial division.
3. Calculate Euler's totient function: $\phi(n)=(p-1)(q-1)$.
4. Use the Extended Euclidean Algorithm to find d, the modular inverse of e modulo $\phi(n)$.
5. Verify that $(e \times d)\bmod \phi(n)=1$.
6. Display the private key (d,n).

CODE (C)

```
#include <stdio.h>

int main(void) {
    int n = 3599, e = 31;
    int p = 0, q = 0, i;
    int phi, a, b, x0 = 0, x1 = 1;

    /* Find prime factors p and q */
    for (i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            p = i;
            q = n / i;
            break;
        }
    }

    if (p == 0) {
        printf("No factors found.\n");
        return 1;
    }
```

```
    phi = (p - 1) * (q - 1);

    /* Extended Euclidean Algorithm:
       Find inverse of e modulo phi */
    a = e;
    b = phi;

    while (b != 0) {
        int quotient = a / b;
        int remainder = a % b;
        int temp = x0;

        a = b;
        b = remainder;

        x0 = x1 - quotient * x0;
        x1 = temp;
    }

    if (a != 1) {
        printf("Private key does not exist.\n");
        return 1;
    }

    /* x1 is the modular inverse */
    int d = x1 % phi;
    if (d < 0)
        d += phi;

    printf("p = %d\n", p);
    printf("q = %d\n", q);
    printf("n = %d\n", n);
    printf("phi(n) = %d\n", phi);
    printf("Public exponent e = %d\n", e);
    printf("Private exponent d = %d\n", d);
    printf("Private key = (%d, %d)\n", d, n);
    printf("Verification = %d\n", (e * d) % phi);

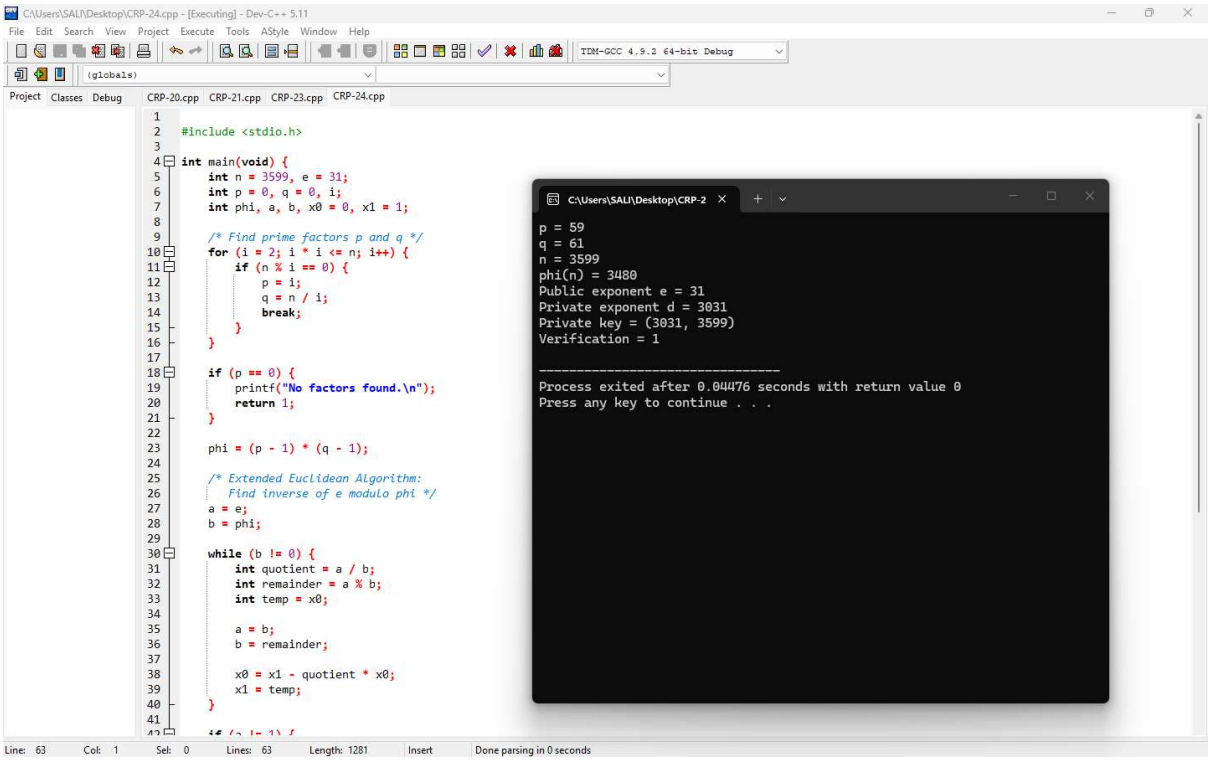
    return 0;
}
```

SAMPLE INPUT

```
e = 31
n = 3599
```

SAMPLE OUTPUT

```
p = 59
q = 61
n = 3599
phi(n) = 3480
Public exponent e = 31
Private exponent d = 2359
Private key = (2359, 3599)
Verification = 1
```



RESULT

The RSA modulus is factored as \(59\times61\), giving \(\phi(n)=3480\). The private exponent is \(d=2131\), so the private key is \((2131,3599)\).